44th AIAA Aerospace Sciences Meeting and Exhibit
9 - 12 January 2006, Reno, Nevada

AIAA 2006-157

Large Eddy Simulation of a confined jet: influence of combustion on the velocity field.

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Today environmental issues play a vital role in the viability of modern, low-emission power plants in which gas turbine combustors need to be operated under lean premixed conditions. However, successful designs require a detailed knowledge of the combustion process. The present study focuses on studying reacting and non-reacting jet. Firstly, an iso-thermal confined jet has been studied using Large Eddy Simulation. The flow is highly turbulent and a precession of the jet core was detected. The comparison of the numerical results and the experimental velocity data measured with Particle Image Velocimetry (PIV) showed an excellent agreement. Secondly, the influence of the combustion on the flow field was investigated. The jet core was found to be lengthened due to the density jump across the flame. The slow jet core precession was not found to be affected by combustion.

Nomenclature

~	=	density weighted space filtering
a	=	ratio between the filter length and the laminar flame thickness
A	=	amplitude of the velocity spectrum
c	=	progress variable
D	=	diffusivity
Dp	=	Diameter of the pipe at the combustor inlet
Δ	=	filter length
PDF	=	Probability Density Function
ρ	=	density
S_L	=	laminar flame speed with respect to the fresh gases
St	=	Strouhal number (frequency*Dp/Inlet bulk velocity)
ω	=	reaction term
X_u	=	variable X in the fresh gases
X_b	=	variable X in the burnt gases
Ξ	=	subgrid scale wrinkling factor
Y_F	=	fuel mass fraction

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I. Introduction

DUE to stricter emission regulations, environmental issues of power generation play an important role in the economic viability of modern power plants. In order to reduce harmful emissions, the current trend is to design gas turbine combustion devices to operate at premixed, extremely fuel lean conditions. Consequently, the peak temperature and NO_X emissions are reduced but combustion instabilities leading to flash back, flame blow out, and high CO/UHC emissions may occur[1]. The challenge therefore is to get an accurate knowledge of the unsteady combustion process in order to design high-efficient, safe, low-emission combustors.

The phenomena are very complex as many physical and chemical scales are involved, therefore turbulent combustion instabilities are a multi-scale 4D problem. Simulation of turbulent combustion is particularly difficult, as the phenomena involved are highly non-linear and unsteady. In addition, a direct full simulation of the problem is beyond our computing capability. However, the recent increase in computational power enables to use Large Eddy Simulation (LES) tools for resolving directly the large scale flame fluctuations and capture the possible instabilities.

In the present paper, LES of a confined jet and jet flame were performed. Turbulent free jets have been studied numerically using LES for about one decade e.g.[2 3 4]. In the initial part of the jet, close to the nozzle exit, the axial velocity along the jet axis remains constant. Further downstream the axial velocity starts to decrease. The decrease corresponds to the end of the jet core. LES studies showed the influence of vortex rings that result from Kelvin-Helmholtz type of instability. They are formed in the shear layer around the jet core and are convected downstream until they merge. These vortex rings contribute to the momentum exchange between the jet and the surrounding. Since vortex pairing is an efficient way of momentum exchange an increase in momentum transfer is visible after pairing. DaSilva and Métais[4] showed that the inflow conditions can trigger vortex pairing. Consequently, one can shorten the vortex core by adjusting the turbulence properties upstream of the expansion.

Turbulent jet flames are simpler to model than swirling flames but are already very complex systems. Additionally to the physical phenomena described in the previous paragraph the thermal expansion due to combustion has to be taken into account. Laboratory experiments related to premixed free jet flames (often called Bunsen burner) using laser techniques are quite common. However, detailed measurements of both velocity and flame properties are much less common. For example Chen et al.[5] measured the velocity and the concentration in a piloted stoichiometric Bunsen-type flame at ambient temperature and pressure. They observed that the thermal expansion decreased the momentum transfer between the core and the surrounding and therefore the jet core became longer. An experimental study at gas turbine relevant operating conditions were performed by Griebel et al.[6 7 8]. He investigated preheated lean premixed jet flames at pressures up to 15 bars.

The present paper uses the setup studied by Griebel et al. [6 7 8] to investigate in detail the influence of combustion on a confined jet flame. Firstly, the LES methodology is presented. In the third section, we present the LES combustion model used for the simulations. The fourth section describes the combustion chamber and the runs performed. The fifth section discusses the accuracy of the LES tool using comparison with experiments and sensitivity analysis. It also includes a comparison between different operating conditions. Finally conclusions are drawn regarding the flame dynamics and the influence of combustion on the jet turbulence.

II. Governing equations & subgrid scale transport modelling

In the present paper, a numerical investigation of a jet and jet flame is considered. In the LES framework, the filtered governing equations are[9]:

$$\frac{\partial \overline{\rho}}{\partial t} + \nabla \cdot \left(\overline{\rho}\,\tilde{u}\right) = 0 \tag{1}$$

$$\frac{\partial \overline{\rho}\,\tilde{u}}{\partial t} + \nabla \cdot \left(\overline{\rho}\,\tilde{u}\,\tilde{u}\right) = -\nabla\,\overline{P} + \nabla \cdot \left(-\overline{\rho\,uu} + \overline{\rho}\,\tilde{u}\,\tilde{u} + \mu \nabla \tilde{u}\right) \tag{2}$$

$$\frac{\partial \overline{\rho}\,\tilde{Y}_i}{\partial t} + \nabla \cdot \left(\overline{\rho}\,\tilde{u}\,\tilde{Y}_i\right) = \nabla \cdot \left(-\overline{\rho\,uY_i} + \overline{\rho}\,\tilde{u}\,\tilde{Y}_i + \rho D_i \nabla \tilde{Y}_i\right) + \overline{\omega}_i \tag{3}$$

$$\frac{\partial \overline{\rho}\,\tilde{T}}{\partial t} + \nabla \cdot \left(\overline{\rho}\,\tilde{u}\,\tilde{T}\right) = \nabla \cdot \left(-\overline{\rho\,uT} + \overline{\rho}\,\tilde{u}\,\tilde{T} + \rho D_T \nabla \tilde{T}\right) + \overline{\omega}_T \tag{4}$$

$$\overline{\rho} = \overline{\left(\frac{P}{RT}\right)} \tag{5}$$

where u is the velocity vector, T the temperature, ρ the density, μ the viscosity, R the specific ideal gas constant, D_i the diffusivity of species i, Y_i the mass fraction of species i and ω_i the reaction term of species i. The bar denotes the spatial filtering operation and the tilde the density weighted spatial filtering:

$$\tilde{Q} = \frac{\overline{\rho Q}}{\overline{\rho}} \tag{6}$$

The spatial filtering corresponds to a "low-pass" filter which removes all the Fourier components that have shorter length scale than the filter size. The filtering operator is linear and is assumed to be commutative with time- and space-derivatives. The filtering operation is not commutative with non-linear terms. Thus, non-linear terms lead to expressions that cannot be expressed in terms of the filtered quantities. These terms are gathered on the right-hand side in the equations above and called subgrid scale (SGS) term. Expressing these SGS terms in terms of filtered quantities is required for "closing" the system. The main role of the SGS terms is to account for the interaction between the resolved and the unresolved scales. In the momentum equations, the SGS terms should account for the dissipative character of turbulence on the small (unresolved) scales as well as for the transfer of energy among the resolved and unresolved scales. Similar SGS properties one may attribute to the SGS in the energy- and the species-transport equations.

A computational grid can support only Fourier components that have longer wavelengths than the grid size. Thus, a dependent variable that is represented on a grid that is used together with a discrete approximation for the derivatives leads to an *implicit* filtering. If no explicit SGS terms are added, then the numerical scheme should account at least for the small scale dissipation. This is attained for any numerically stable scheme. However, too dissipative numerical schemes are highly in-appropriate for LES since in addition to dissipation on small scales, they may be also dissipative on the larger scales. This effect can be avoided by choosing appropriate (higher order) discretizations. In addition, the spatial resolution has to be fine enough, of the order of magnitude of the Taylor micro-scale. With such a resolution, the energy transfer among the scales is dissipation independent and therefore the numerical scheme may act implicitly as a SGS model. However, one should keep in mind that this implicit SGS approach requires a better resolution than a corresponding LES which uses advanced SGS models. Nevertheless, even with explicit SGS (such as the dynamic model), the spatial resolution has to be adequate (usually may be coarser by a factor of at most 2 compared to the implicit SGS approach). In the present paper, we use a third-order upwind scheme[10]. Using the same scheme, Revstedt et al..[11] showed that traditional SGS models do not perform significantly better than the implicit approach for simulations of a turbulent jet. Therefore, the implicit approach will be used in this paper for closing the filtered momentum transport equations.

III. Filtered flamelet model for premixed combustion

The set of equations describing the fuel oxidation contains the transport equations for the considered chemical species and for the enthalpy. The complete set of equations detailing the oxidation of natural gas is far too big to be used for numerical simulations and a reduced formulation is needed. In the present approach, the flamelet assumption allows a simplifying coupling of turbulence and chemistry[9 12]. The combustion process is described by a single parameter called progress variable *c*. *c* is related to the advancement of the reaction. *c* is defined to be 0 in the unburnt gases and 1 in the burnt gases. In this paper, c is based on the fuel mass fraction Y_F. We normalize the fuel mass fraction and rewrite equation (3) to be:

$$\frac{\partial \overline{\rho}\tilde{c}}{\partial t} + \nabla \cdot \left(\overline{\rho}\tilde{u}\tilde{c}\right) = \nabla \cdot \left(-\overline{\rho u c} + \overline{\rho}\tilde{u}\tilde{c}\right) + \nabla \cdot \left(\overline{\rho D_{th} \nabla c}\right) + \overline{\omega_c} \tag{7}$$

The left-hand side of equation (7) contains the unsteady and convective terms. The right hand side contains the diffusion, the reaction and the subgrid scale transport.

A major difficulty in LES of reacting flows is that chemical reactions occur at scales much smaller than the filter length and therefore are not resolved and require modeling. In addition, the sum of the three terms on the right hand side of equation (7) should model the premixed flame propagation accurately. In the present paper, a premixed flamelet formulation used by Duwig and Fuchs[12] has been used to close the filtered *c*-equation. This approach

consists of modeling the three unclosed terms of equation (7) by a reactive-diffusive balance ensuring correct flame propagation. The closed c-equation reads:

$$\frac{\partial \overline{\rho}\tilde{c}}{\partial t} + \nabla \cdot \left(\overline{\rho}\tilde{u}\tilde{c}\right) = \frac{\rho_u S_L \Xi \Delta}{a} \nabla^2 \tilde{c} + \rho_u S_L \Xi \frac{1}{\Delta} \Pi_c(\tilde{c}, a) \tag{8}$$

where Π_c is a production term and a is a non-dimensional number comparing the filter size Δ and the laminar flame thickness δ, $a=\Delta/\delta$. Duwig and Fuchs[12] derived an expression for Π_c in the flamelet regime, Figure 1 shows the production term mapped in a c, a space.

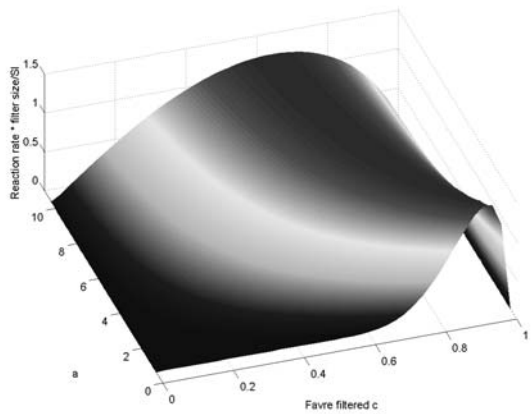

Figure 1: Normalized production term Π_C in a Favre filtered c / a space

IV. Computational tools and burner setup

A. The LES tool

The present formulation has been implemented into a high-order finite difference code that solves the incompressible Navier-Stokes equations with variable density on Cartesian grids. The code is third-order accurate for the convective terms and forth-order accurate for the other terms. A second order finite difference scheme is used for time discretization. Locally refined grids can be employed in regions with large gradients. Multi-grid iterations are used to solve the system of equations. More details can be found elsewhere[13]. Using Cartesian finite differences yield easily coded, fast and accurate results. These advantages make the approach most suitable for LES of reacting flows.

B. The burner

Non-reacting and reacting flows in a dump combustor where studied. The geometry corresponds to an experimental high-pressure test rig located at the Paul Scherrer Institut (Villigen PSI, Switzerland). The generic, high-pressure combustor consists of an inlet pipe diameter of $Dp=25\cdot10^{-3}$ m discharging in a coaxial pipe of diameter 3Dp. The flame is stabilized aerodynamically by the recirculation of hot combustion gases induced by the geometry with sudden expansion. The combustor is equipped with high pressure windows to provide an optical access to the entire length of the combustor chamber. The turbulence intensity (u') and the integral length scale at the inlet can be varied by using different turbulence grids (defined by hole diameter and blockage ratio) or by mounting these grids at different axial positions within the combustor inlet section. In this study a perforated plate with hexagonally located 3 mm (0.12Dp) holes and a blockage ratio of 50 % was used. The grid was located at the axial distance of 1.2Dp upstream of the expansion. The turbulent flow field was measured with PIV in the isothermal flow field (Griebel et al.[7]). Detailed experimental data are available for the present jet flame[6 8].

The computational domain starts 1.2Dp upstream of the expansion and ends 12.4Dp downstream of the expansion A steady Neumann inflow boundary conditions is used i.e. the inflow is modeled as a collection of small jets. Modeling of the incoming turbulence is a difficult but important task for LES computations. Steady inflow

conditions ensure a correct incoming mass flow but do not mimic many features of turbulence. However, the strong shear imposed at the inlet generates some turbulence. As we will see later, the turbulent fluctuations level at the expansion is predicted correctly. Despite of injecting no turbulence in the domain, we obtain an accurate turbulent field upstream of the expansion. Zero gradient outflow boundary conditions are used.

The computational domain includes the whole 3D geometry. Three different computational grids were tested. The coarser grid consists of 1000000 points giving a mesh spacing of ~~$1\cdot10^{-3}$ m in the expansion region. The medium size grid contained 1500000 points with a refined region close to the expansion with mesh spacing of ~$0.6\cdot10^{-3}$ m. Finally the finer grid contains 2500000 points. It is similar than the medium grid but the locally refined region includes a larger part of the domain. The time step was computed using a CFL number ~0.2.

In the non-reacting case, the inflow bulk velocity was set to 40 m/s giving a Reynolds number of 64000. The operating temperature and pressure where respectively 293K and 101325 Pa.

In the reacting cases, the bulk velocity was kept constant to 40 m/s. A methane/air mixture of equivalence ratio 0.5 was used. The fresh gas temperature was set to 673 K. The operating pressure is 101325 Pa corresponding to a Reynolds number of 24000. The Karlowitz number is ~5. The flame is in the so-called extended flamelet regime but not in the flamelet regime. However, Duwig and Fuchs[12] showed that the filtered flamelet model presented in section III can be also used for accurate simulations within the extended flamelet regime.

For both reacting and non-reacting cases, the velocity quantities are dimensionless by scaling with the bulk velocity of 40 m/s.

V. Results and discussion

A. Non-reacting case

Firstly, the influence of the computational grid density on the results was investigated. Figure 1 shows the normalized mean axial velocity profiles at 5 different axial locations in the combustor. The jet penetrates into the combustor and spreads creating a recirculation zone in the outer region. All three grids are capturing the mean flow pattern properly and predict similarly the spreading rate and the recirculation zone length/strength. Figure 2 shows the root-mean-square of the axial velocity. The fluctuation levels predicted by all three grids are very similar both in magnitude and in shape. In principle one could conclude that all these three grid are suitable for simulating the non reacting jet. However, the present study includes also simulations of reacting jet at higher Reynolds number (i.e. cases that require higher resolution). Consequently, we decide to use the medium grid for the rest of the study.

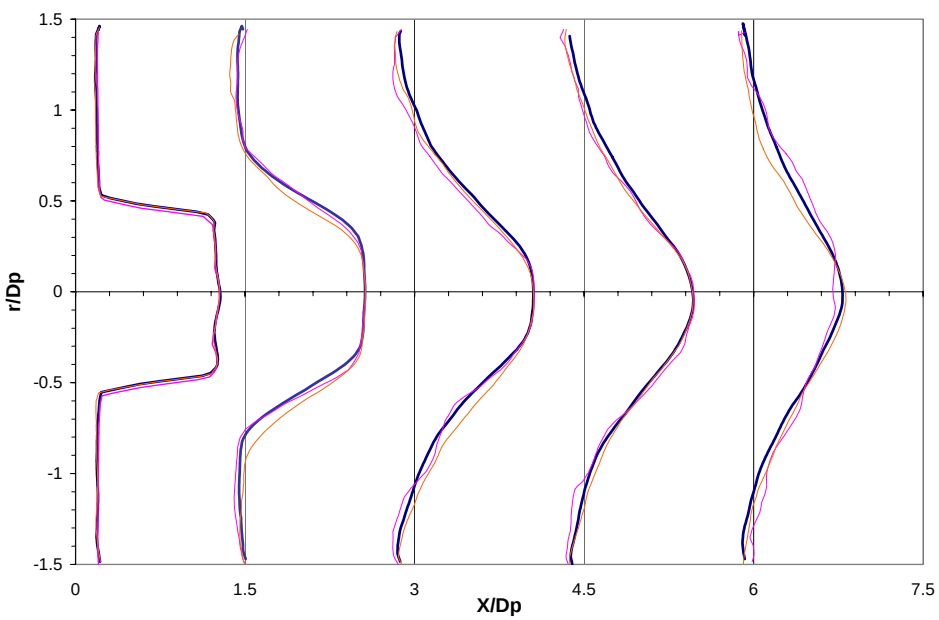

Figure 1: Radial profiles of the normalized mean axial velocity u_{mean}/u_{bulk} at different axial positions (x/Dp=0.2, 1.5, 3, 4.5, 6) . Red: fine grid; light blue: coarse grid; dark blue: medium grid. One unit on the x-axis corresponds to 1 nomalized velocity unit.

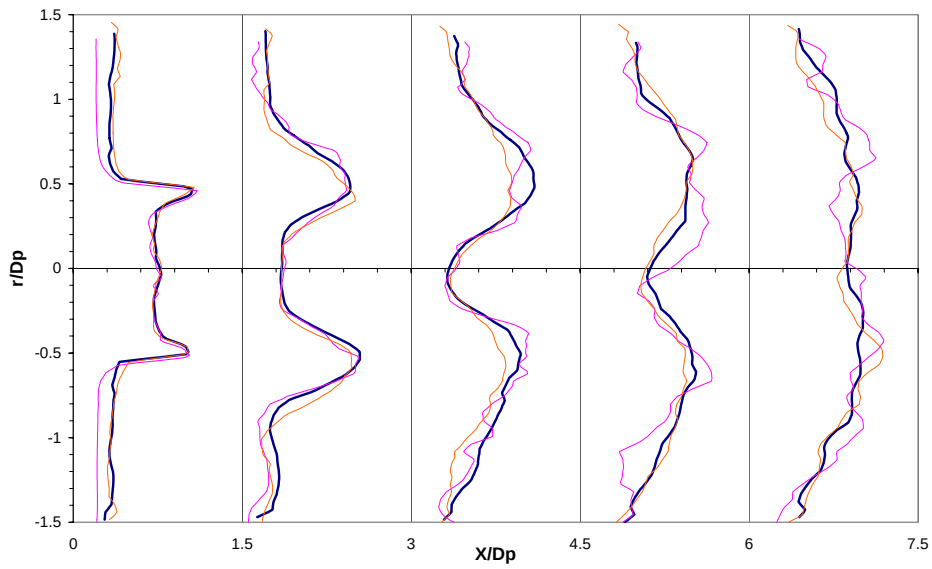

Figure 2: Radial profiles of the normalized axial velocity RMS u_{RMS}/u_{bulk} at different axial positions. The legend is the same than on figure 1. One unit on the x-axis corresponds to 0.2 nomalized velocity unit.

The predictions performed with the medium grid were compared to available PIV measurements[7]. Figure 3 shows radial profiles of the axial velocity. The LES predictions match very well the PIV measurements. The jet spreading rates as well as the outer recirculation zone are well captured by LES. The discrepancy between the predictions and the experiments is below 5 % which is in the order of the estimated accuracy of the PIV results.
Figure 4 shows the radial profiles of the RMS of the normalized axial velocity. In general the predictions agree very well with the measurements. At the axial positions close to the combustor inlet and at a radius of r/Dp=0.5, the RMS values are very high and the profiles show steep gradients. In the shear layer, turbulence is generated due to high velocity gradients between the jet core and the recirculation zone leading to high velocity fluctuations in this region. In the outer region (recirculation zone) the fluctuations are low. In the core of the jet, the fluctuations are almost constant and the fluctuation level is close to 15% of the bulk velocity. Since the core region is not influenced by strong shear stresses, the fluctuations originate upstream i.e. are generated by the turbulence grid and are convected downstream by the flow. Figure 4 shows that the fluctuation levels predicted by LES are in good agreement with the experimental data. This validates our modeling of the inflow boundary condition.

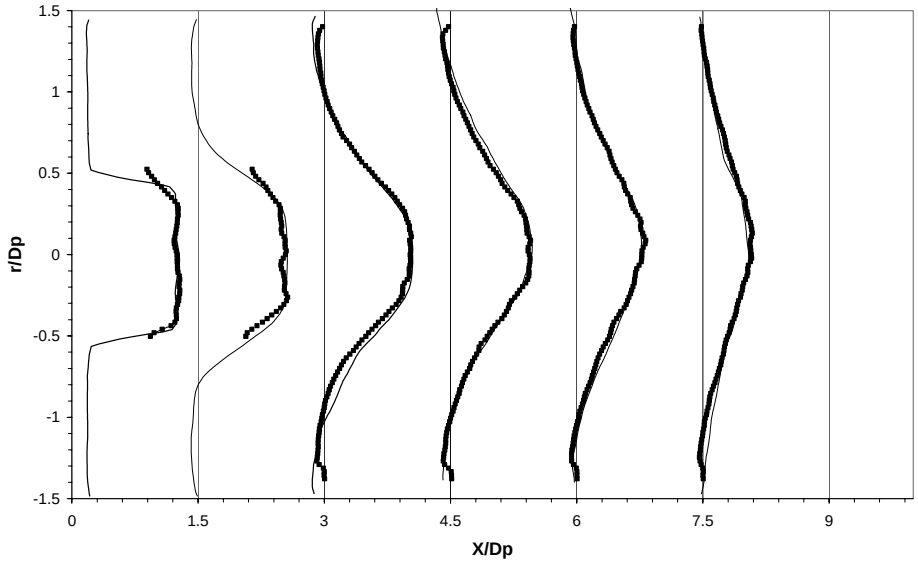

Figure 3: Radial profiles of the normalized mean axial velocity at different axial positions (x/Dp=0.2, 1.5, 3, 4.5, 6, 7.5) . The lines shows the LES predictions while the squares represents the PIV measurements [7]. One unit on the x-axis corresponds to 1 nomalized velocity unit.

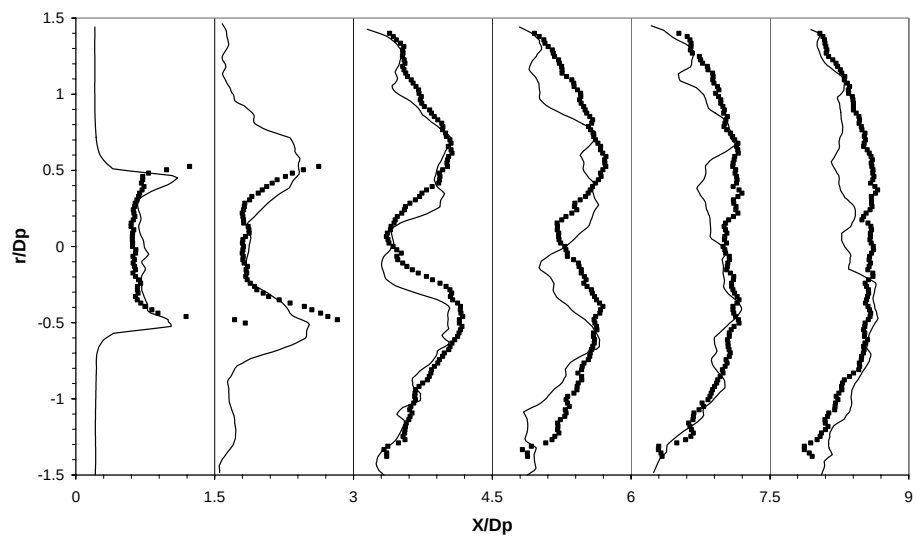

Figure 4: Radial profiles of the normalized RMS of the mean axial velocity at different axial position (x/Dp=0.2, 1.5, 3, 4.5, 6, 7.5) . The lines shows the LES predictions while the squares represents the PIV measurements [7]. One unit on the x-axis corresponds to 0.2 nomalized velocity unit.

Figure 5 shows a snapshot of the jet core. The core surface is wrinkled by the turbulent structures. Close to the expansion, the vertical structures are relatively small and located in the shear-layer. They are responsible for the peaks seen on figure 2 and 4. Further downstream, the size of the vertical structures and the characteristic length of the wrinkling increase. Therefore the momentum exchange between the jet core and the surrounding fluid is amplified and results in an increase of the jet spreading rate as seen in figure 3. Figure 6 shows that the fluctuating kinetic energy is concentrated around a finite number of characteristic frequencies. A typical shear-layer instability is located for St~0.3. It is characteristic for the turbulence generated in the shear layer and corresponds to the structures visualized on figure 5. In addition two peaks at St=0.035 and St=0.07 are visible in figure 6. It is believed that these peaks correspond to a slow precession of the jet core around the combustor axis. Since the period of this phenomenon is very slow, a phase averaging procedure is not feasible.

The presence of a slow large scale mode is confirmed by studying the axial velocity PDF. Figure 7 shows two points located at x/Dp=3. The simulation results are shown on the left hand side whereas the experimental results are listed on the right hand side. Firstly, it can be concluded that the LES computation capture very well the shape of the measured velocity PDF indicating that the present computation does not only capture the averaged quantities but also the nature and magnitude of the fluctuations. The first point is located at the centerline in the jet core. The width of the PDF is very narrow and the PDF has a Gaussian shape, indicating that the instantaneous velocity fluctuations are simply distributed around the mean value. The fluctuations at this location originate from the grid and are convected downstream. The second location lies within the shear-layer close to the end of the jet core. The shape of the PDF is rather complex with three peaks. The broad width of the PDF shows also a large fluctuation amplitude of the velocity. The particular shape can be explained considering the slow precession of the jet core. During the precession, the shear-layer will move and location 2 will be sometimes in the core of the jet (relatively high velocity) and sometimes outside of the jet core (relatively low velocity). This particular motion results in two distinct states. The superposition of the two states results in the broad PDF with three peaks as presented in figure 7.

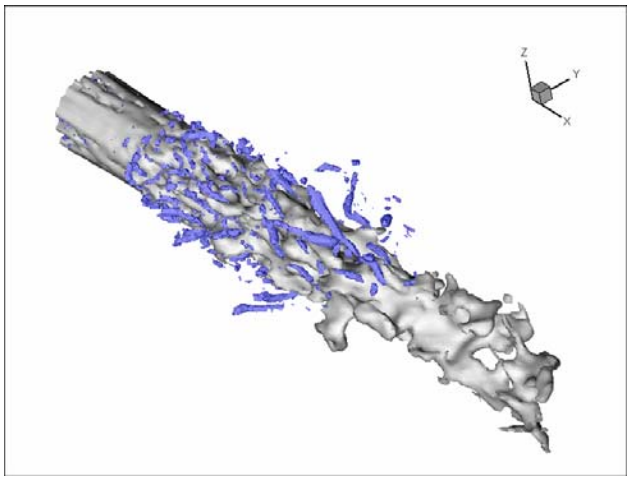

Figure 5: Grey: snapshot of the axial velocity level (U/Ubulk=0.75). Blue: snapshot of the vorticity tubes visualized as iso-levels of λ_2[14].

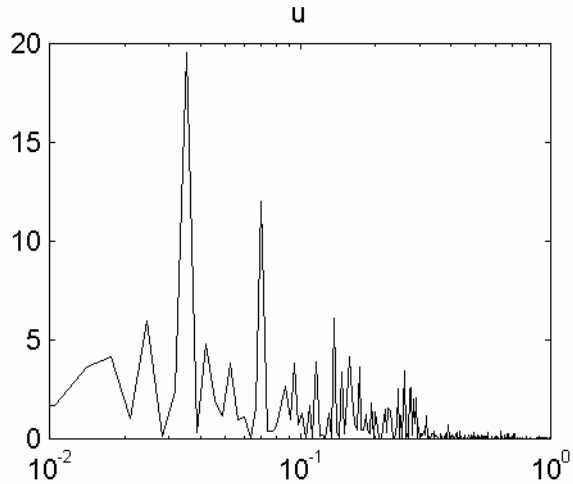

Figure 6: Power spectra of the axial velocity fluctuation vs Strouhal number (=f*Dp/Ubulk). The sampling point is located in the jet shear-layer (x/Dp=3, r/Dp=1).

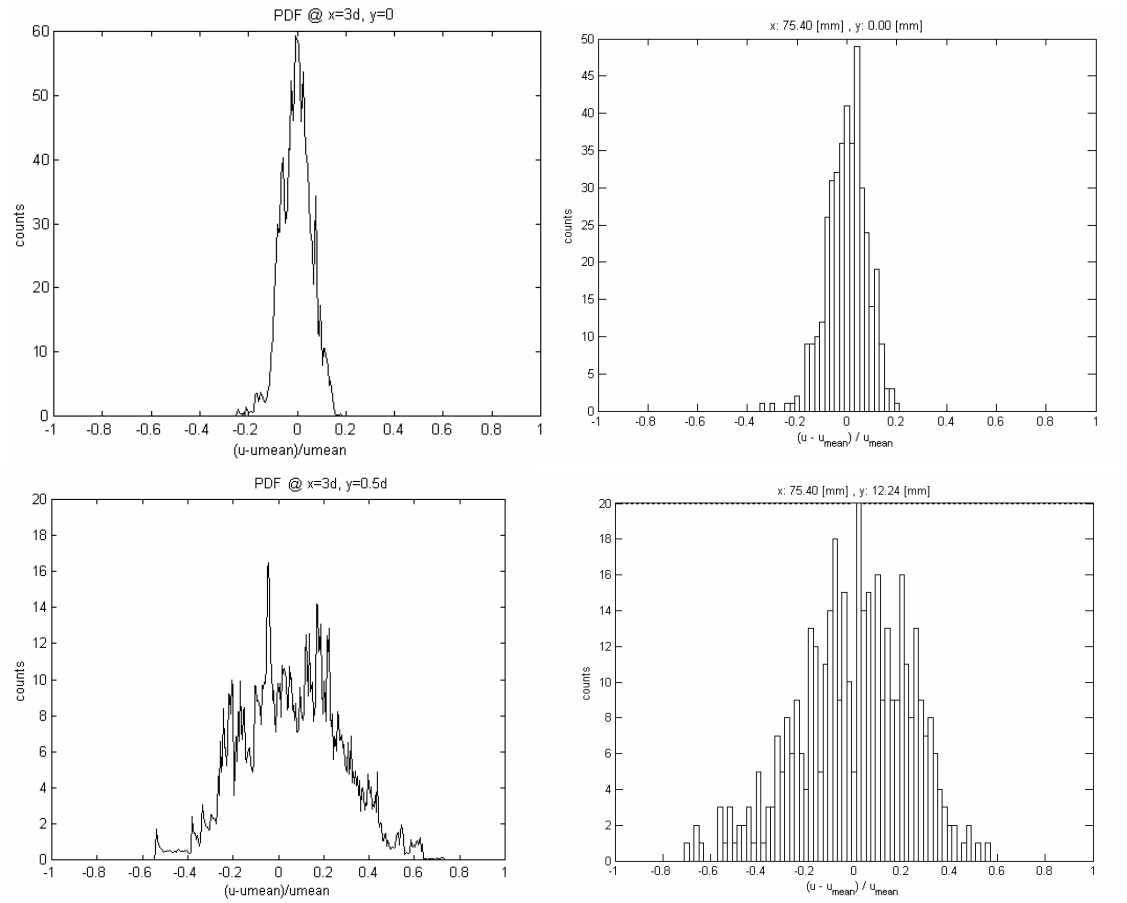

Figure 7: Probability density function (PDF) of the reduced axial velocity at two different locations: Up (x/Dp=3, r/Dp=0.) and Down (x/Dp=3, r/Dp=0.5). LES predictions are shown on the left and experimental data are presented in the right part.

B. Influence of combustion on the flow field

Firstly, we consider the reacting case at atmospheric pressure. Figure 8 left shows the instantaneous flame front wrinkled by vortical structures. The flame surface is enlarged and the turbulent flame speed is increased by turbulence. Close to the expansion, the flame wrinkling is due to small scales. Further downstream, large scale wrinkling is seen and the flame shape changes from quasi-cylindrical to a spiral-type. Figure 8 right shows the vortex tubes. Close to the expansion, the vortex tubes are relatively small and do not modify significantly the global flame pattern. Further downstream, large scale wrinkling occurs as large vortical structures appear. The vorticity dynamics is similar to what was observed in the non-reacting case.

Figure 9 shows a comparison of the axial velocity field for a reacting and a non-reacting case. At the exit of the nozzle, the profiles are identical. Farther downstream, the reacting jet core is slightly longer and wider. The momentum transport from the jet to the recirculation zone is therefore lower in the reacting case. This phenomenon is well known as ‘counter-gradient transport’ (CGT). It describes the effect of a thermal expansion within a shear-layer. The differential acceleration of the “cold” gases (the jet) and the combustion products (recirculation zone) lowers the entrainment of the jet. Consequently in this particular setup CGT slightly decreases the axial expansion of the jet core. This phenomenon has also been observed in free jets[5].

Figure 10 left shows the axial velocity PDF in the shear-layer at an axial distance of 3Dp from the expansion. The shape of the PDF is not Gaussian and again 3 peaks are visible. Comparing this PDF with the corresponding one in figure 7, it is somewhat obvious that the peak at the negative velocity is dominant in the reacting case but not in the cold case. The negative velocity peak corresponds to the moments where hot gases are found at this location.

Hence, thermal expansion increases the back flow velocity. Figure 10 right shows the frequency spectra. As for the non-reacting case, low frequency motions are observed, i.e. jet core precession. However, the characteristic frequencies of the two peaks are shifted to St~0.025 and St~0.05 indicating that the thermal expansion slows down the precession of the jet core.

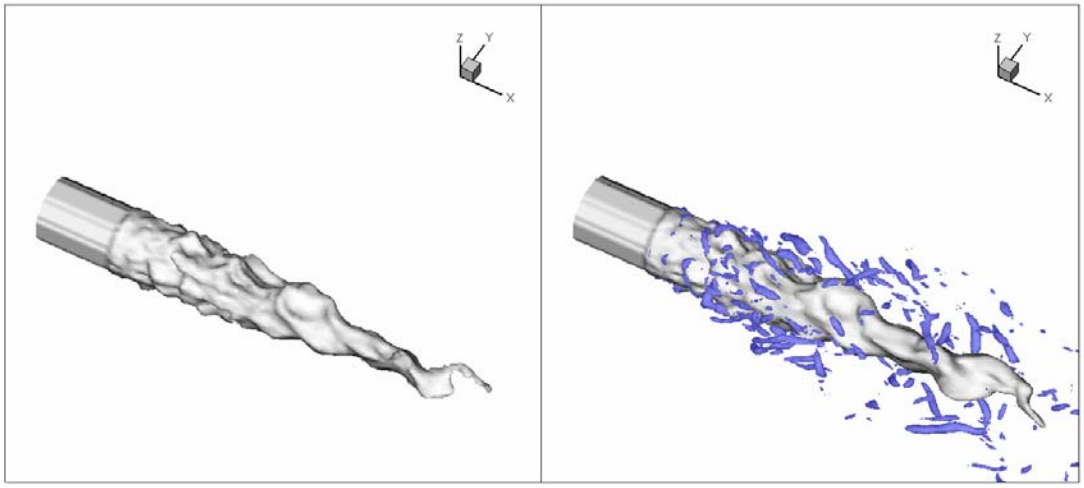

Figure 8: Set of snapshots of the flame front (iso-surface c=0.5). Left: 3D surface of the flame front attached to the inflow pipe. Right: the flame front is wrinkled by the vortex tubes (blue surfaces[13]).

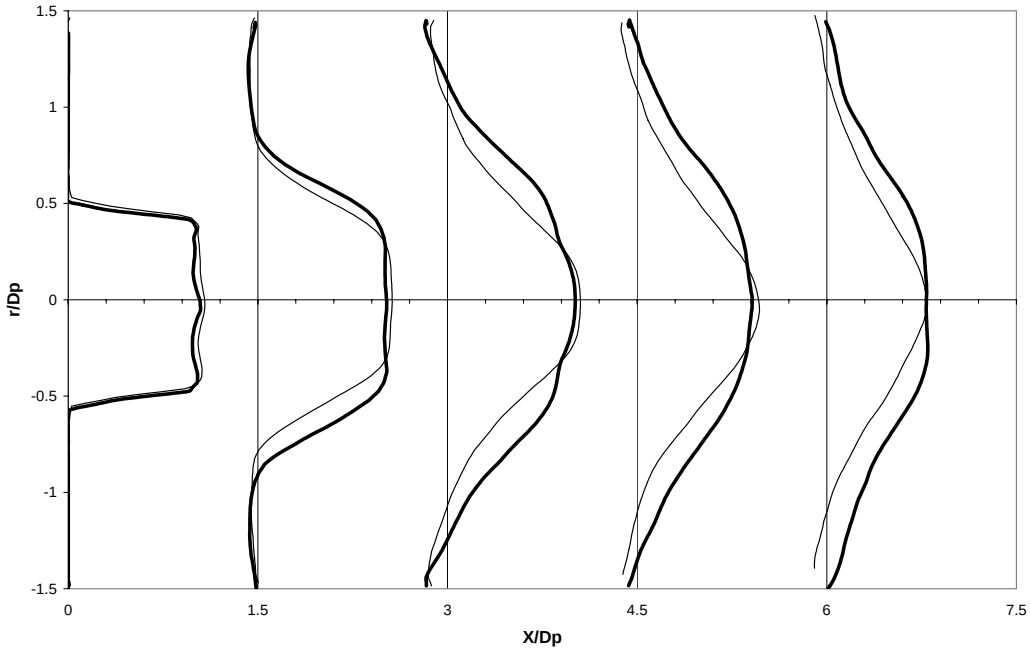

Figure 9: Radial profiles of the normalized mean axial velocity at different axial positions (x/Dp=0.2, 1.5, 3, 4.5, 6). The thin line shows the cold LES predictions while the thick line shows the results of the reacting case at 1 bar. One unit on the x-axis corresponds to 1 nomalized velocity unit.

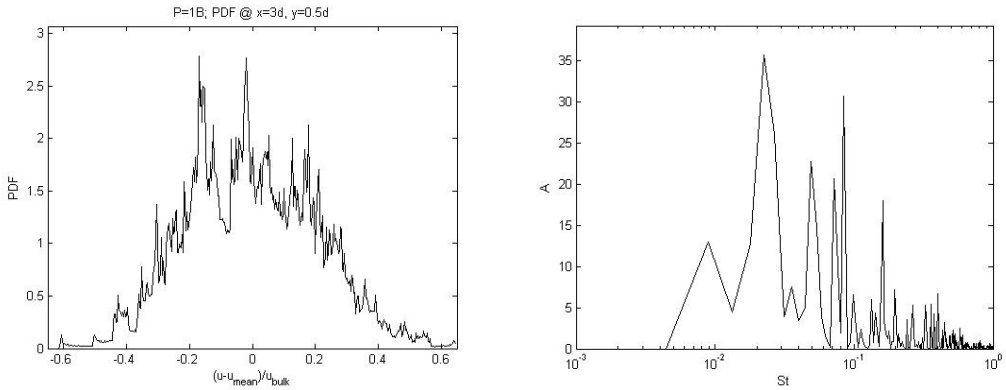

Figure 10: Left: Axial velocity PDF at (x/Dp=3, r/Dp=0.5).
Right: Axial velocity frequency spectra at (x/Dp=3, r/Dp=0.5) vs. Strouhal number.

VI. Summary

The present paper reports a numerical investigation of a turbulent confined jet. Numerical simulations of non-reacting and reacting conditions were performed. The reacting conditions correspond to a lean premixed flame relevant for gas turbine applications. Large Eddy Simulation was used to investigate the flow and was supplemented with a LES flamelet model for capturing the flame dynamics. The velocity results of the non-reacting case were averaged and compared to experimental PIV results. The agreement was excellent for both the mean and the RMS values. The numerical simulation showed that the jet core exhibits a slow precession.

The simulations of the reacting cases showed that the thermal expansion across the flame front lowers the momentum exchange between the jet core and the recirculation zone. Consequently, the jet core was found to be slightly longer and broader in the reacting case. However, this change in the aerodynamics did not suppress the slow precession of the jet core but shifted its frequency. One may argue that the preheating of the fuel/air mixture reduced the density jump across the flame front and consequently lowers its effect on the flow field. It highlights the relevance of isothermal experiments/simulations for understanding the flow dynamics in combustors and shows the possibility to extend this understanding to reacting cases.

Acknowledgments

This work was supported by the Swedish Energy Authority (STEM) within the framework, ‘termiska processer för el-producktion’. The computations were run on LUNAC and HPC2N facilities within the allocation program SNAC.

The experimental part of this work was financially supported by the Swiss Federal Office of Energy (BFE) which is gratefully acknowledged.

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